

Access CityU HK

A Crowdsourced Accessibility Management
Platform for Inclusive Campuses



Agenda

01

Problem & Context

Analyzing core challenges, market dynamics, and stakeholder requirements to establish the project's foundational backdrop.

04

IT Integration Strategy

Detailing the roadmap for system interoperability, data migration, and seamless alignment with existing IT infrastructure.

02

Project Objectives

Defining clear, measurable goals and success metrics that align with overarching business vision and strategic priorities.

05

Implementation Plan

Mapping the phased rollout, resource allocation, risk mitigation strategies, and key milestones for execution.

03

Solution Framework

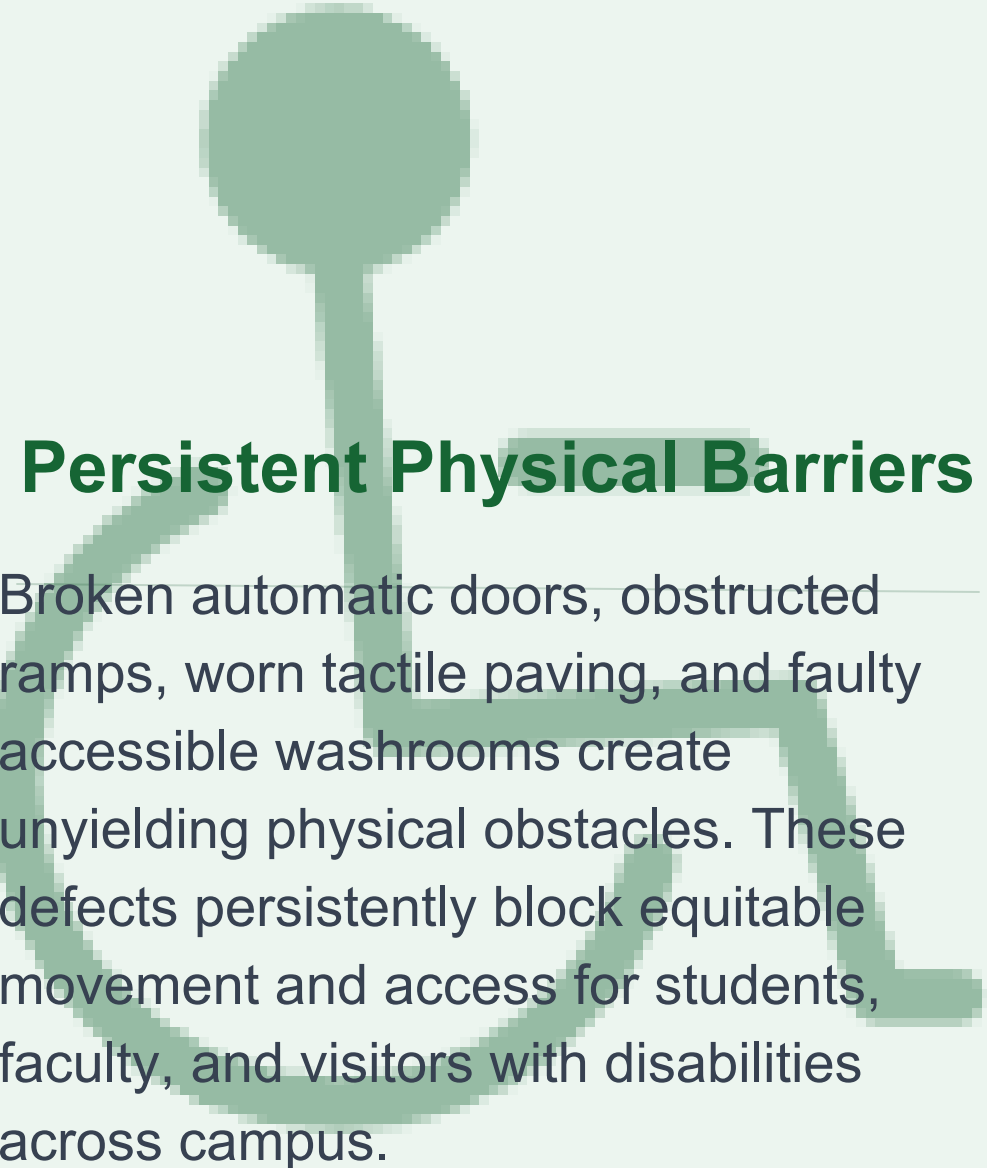
Outlining the architectural blueprint, key functionalities, and technological pillars that underpin the proposed solution.

06

Expected Impact & SDGs

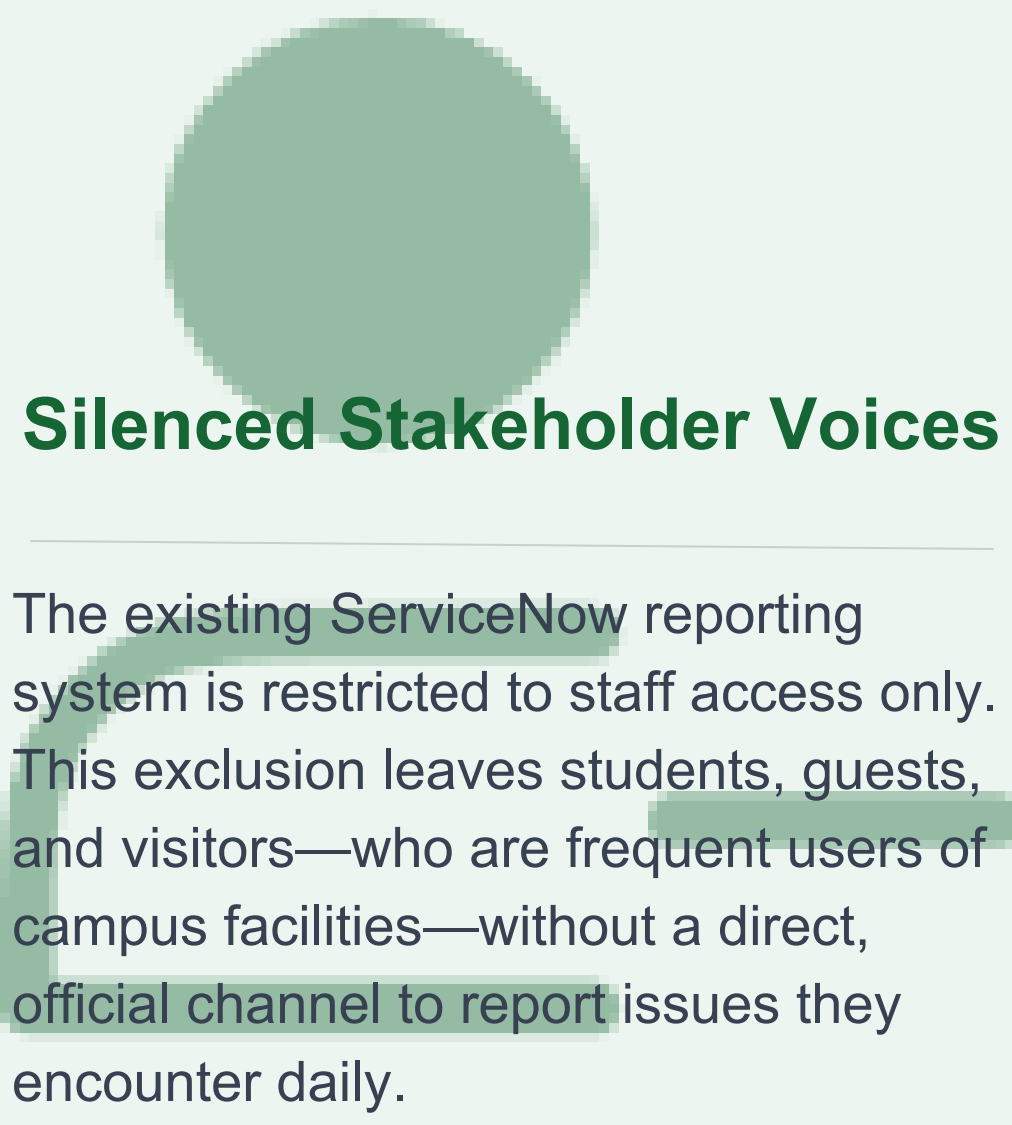
Evaluating economic, social, and environmental outcomes, including alignment with UN Sustainable Development Goals.

The Problem: Inconsistent Accessibility Maintenance



Persistent Physical Barriers

Broken automatic doors, obstructed ramps, worn tactile paving, and faulty accessible washrooms create unyielding physical obstacles. These defects persistently block equitable movement and access for students, faculty, and visitors with disabilities across campus.



Silenced Stakeholder Voices

The existing ServiceNow reporting system is restricted to staff access only. This exclusion leaves students, guests, and visitors—who are frequent users of campus facilities—without a direct, official channel to report issues they encounter daily.



Escalating Safety Hazards

Delayed reporting of all the facility issues allows minor defects to go unaddressed. A stuck door or uneven pavement quickly evolves from a minor inconvenience into a serious safety risk, endangering the well-being of those who depend on accessible infrastructure.

Key Pain Points

For Students

No direct reporting channel

Absence of a dedicated platform creates initial barriers, leaving students unsure how to escalate accessibility issues effectively.

Unclear repair timeline

No visibility into when fixes will be completed disrupts daily living, academic schedules, and overall student well-being.

Repeated follow-up emails

Endless, time-consuming follow-ups with no tangible updates drain energy and focus away from academic responsibilities.

Safety & dignity compromised

Unresolved issues create physical hazards and emotional distress, undermining the student's sense of autonomy and belonging.

For FMO / University

Duplicate submissions

Multiple entries for the same issue flood systems, causing data clutter and wasting staff time sorting through redundant tickets.

Manual data re-entry

Staff waste hours transferring data between systems, increasing the risk of human error and reducing time for actual problem-solving.

No predictive maintenance data

Lack of historical trend analysis means the university cannot anticipate failures, leading to more frequent and costly breakdowns.

Reactive rather than proactive

The team is stuck in a cycle of crisis response instead of strategic planning to improve campus accessibility infrastructure.

Impact: These systemic inefficiencies directly result in eroded safety, independence, and dignity for students with mobility, visual, and neurodiverse needs, creating a campus environment that fails to uphold its commitment to universal accessibility.

Our Vision

Transform accessibility from a passive feature into an actively monitored, student-empowered system that ensures equitable access to campus resources for all.

Unified QR Tracking

Implement a seamless QR code system for end-to-end asset visibility, ensuring every device and resource is accountable and traceable across campus.

Real-time Maintenance

Transparent dashboards provide live status updates on equipment health, enabling rapid response to issues and minimizing downtime for critical tools.

AI Predictive Insights

Leverage machine learning to forecast potential failures, shifting from reactive fixes to proactive care that optimizes resource longevity.

Cross-Dept Synergy

Break down administrative silos to create a unified ecosystem where IT, faculty, and students collaborate to enhance accessibility and usability.

Solution Framework — 4 Stages

01

QR Asset Deployment Student Audit Framework

Deploy intelligent QR code asset tags to achieve unique identification and binding for each campus asset.

Establish an asset lifecycle management system to enable dynamic tracking from procurement and utilization to disposal.

02

Frontline Reporting Backend AI Analytics

Enable frontline users to report incidents instantly via mobile devices, with real-time data aggregation across platforms to break down information silos.

Real-time visualization of key metrics empowers data-driven decision-making, while backend AI models drive intelligent anomaly detection, trend forecasting, and root cause analysis.

03

Campus Outreach Inclusivity Awareness

Carry out diversified campus promotion activities covering all teachers and students to raise awareness of services and boost participation

Build a collaborative feedback loop to refine services through ongoing community input and foster a stronger sense of campus unity.

04

FMO ServiceNow Integration

Enable automated work order workflows with smart dispatching, reducing human touchpoints and accelerating response times.

Build a centralized service portal for end-to-end management, seamlessly integrating service requests, handling, and feedback.

Stage 1 — Universal QR Asset & Audit Framework

01. Campus QR Asset Mapping

- Baseline Inventory:** Conducting a full quantitative spatial audit to map every physical asset across the campus landscape.
- Digital Identity:** Assigning a unique Unified Asset Code (UAC) to each item for permanent traceability.
- Durable Deployment:** Installing weather-resistant acrylic QR labels for long-term outdoor/indoor readability.

02. Student Audit Drives

- Collaborative Effort:** Partnering with the University Sustainability Club to engage the student community actively.
- Compliance Focus:** Using standardized checklists aligned with ADA & ISO 21542 accessibility standards.
- Inclusive Participation:** Opening the audit process to all CityUHK students to foster ownership.

03. Low-Friction Reporting

- Instant Access:** A simple QR scan launches a mobile browser form, no app download required.
- Structured Data:** Pre-categorized dropdowns mirror FMO's taxonomy for data consistency.
- Auto-Context:** Geotags, timestamps, and asset IDs are automatically attached to every report.

Stage 2 — Reporting & Predictive Analytics

01. Frontend: Instant Transactional Reporting

Sub-30s Defect Submission

Ultra-fast form workflow minimizes downtime, ensuring issues are logged immediately as they are identified in the field.

Structured Data Inputs

Eliminates unstructured free text with predefined dropdowns, ensuring data consistency for reliable analytics.

Duplicate Ticket Prevention

Real-time lookup checks for existing cases during submission to avoid redundant work and streamline resolution.

02. Backend: Secure Admin Analytics Dashboard

Power BI Dashboard (FMO Exclusive)

Role-based secure access to real-time fleet health and operational performance metrics.

AI on Structured Historical Data

Machine learning models analyze historical trends to predict future component failures.

Hazard Frequency & Pattern ID

Quantifies failure rates and identifies recurring patterns to address systemic root causes.

High-Risk Zone Heatmaps

Visualizes geographic clusters of defects to optimize inspection routes and resource allocation.

Capital Budget Allocation Support

Data-driven insights prioritize high-impact maintenance projects for optimal ROI.

Proactive Maintenance Triggers

Alerts for potential failures based on threshold breaches in defect frequency.

Stage 3 — Campus Outreach & Awareness

01 / Awareness Week & Promotion

Podium Exhibition Spaces

High-visibility booths across campus for full campaign rollout, ensuring maximum exposure to the student body during peak hours.

DIY Sustainability Workshops

Collaborative sessions with the Sustainability Club on upcycling, zero-waste living, and inclusive green practices for all students.

On-Campus Interactive Demos

Hands-on product trials and live impact simulations to turn passive awareness into active engagement with sustainable living.

02 / Transparent Communication

Audit Summaries

Monthly sustainability audit results published in newsletters to keep the community informed.

Public Resolution Rates

Openly shared ticket resolution metrics to build accountability and trust.

Data Dissemination

Impact data displayed on noticeboards and amplified via social media channels.

Ownership Culture

Empowering students to lead, fostering long-term commitment to campus sustainability.

IT Integration Strategy — IT Department Recommendations

CORE PRINCIPLE: Focus on Standalone Application First — Validate stability, security, and business value of individual systems before pursuing complex cross-functional integrations.

Recommended approach per Central IT guidance: Build independently to minimize risk and demonstrate value; integrate deeper only after standalone success and clear business justification.

Email-to-Ticket

Automate ticket creation from incoming support emails to eliminate manual data entry. This streamlines workflow, ensures zero request leakage, and significantly boosts team response efficiency.

API Roadmap Sync

Establish cross-functional workshops to define API standards, version control protocols, and long-term integration milestones. Aligns technical capabilities with business needs for scalable growth.

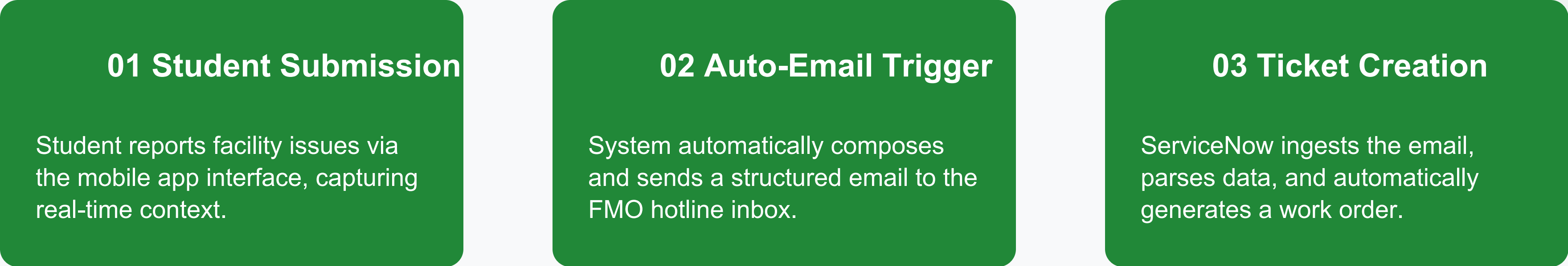
Security & Governance

Enforce strict API rate limits, OAuth2.0 authentication, and end-to-end encryption. Protects sensitive data, prevents system overload, and ensures compliance with data privacy regulations.

Asset QR Mapping

Deploy scannable QR codes on all hardware assets for real-time inventory visibility. Enables automated check-in/out processes and accelerates troubleshooting for on-site IT support.

Email-to-Ticket Solution — Core Integration Mechanism



Building Location Precise campus coordinates, room number, and floor identifier for quick dispatch.	Defect Details Standardized using FMO’s official taxonomy for consistent categorization.	Reporter Notes Contextual user observations and specific issue descriptions.	Asset QR Data Direct link to asset ID, maintenance history, and manufacturer info.
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QR Codes & Asset Mapping Strategy

Our Database (Source System)

Centralized Inventory: Asset records stored securely in a dedicated app database with full ownership.

Internal QR Management: QR codes generated in-house, uniquely mapped to each asset's digital profile.

Data Enrichment: Critical asset details (location, specs, history) auto-appended to outputs.

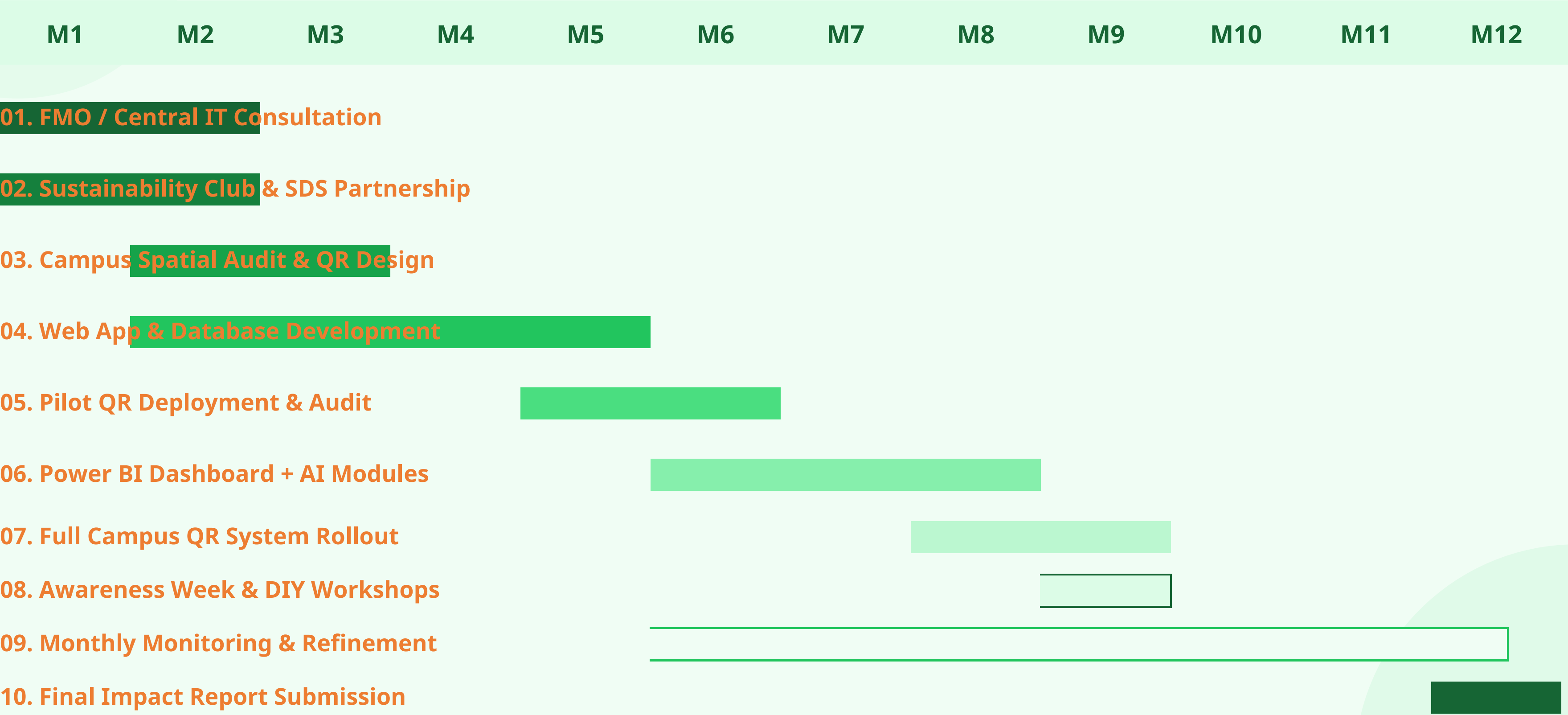
Schema Control: Complete flexibility to modify data structure without external dependencies.

**Email as
Integration Layer**

Existed FMO's sytem

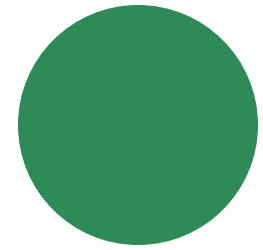
Key Outcome: This strategy preserves the autonomy and standalone nature of our asset management system while ensuring the FMO team receives complete, actionable asset context.

12-Month Implementation Timeline



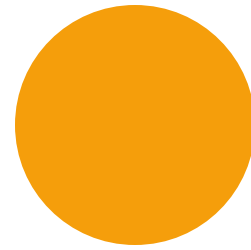
Sustainable Development Goals Alignment

Aligned with 5 UN SDGs — building inclusive, resilient campus communities



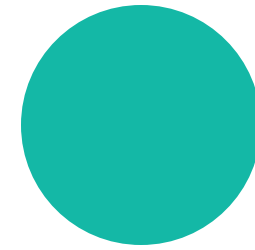
SDG 3: Health & Well-being

Championing holistic wellness, mental health support systems, and accessible healthcare to nurture a thriving campus community.



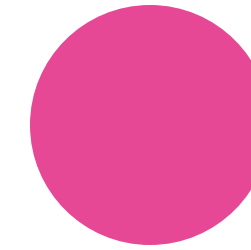
SDG 9: Innovation & Infra

Investing in green tech labs, smart campus infrastructure, and fostering a culture of innovation for sustainable solutions.



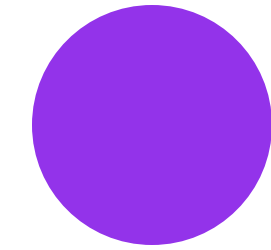
SDG 10: Reduced Inequalities

Promoting equal access to education, scholarships, and resources to eliminate socioeconomic barriers across the campus.



SDG 11: Sustainable Cities

Designing eco-friendly urban spaces, zero-waste systems, and resilient infrastructure for campus living.



SDG 17: Global Partnerships

Collaborating with NGOs, universities, and global networks to amplify impact and share best practices.

By aligning our campus strategy with these UN SDGs, we commit to fostering a legacy of sustainability, equity, and innovation that empowers our community to drive positive global change.

Expected Outcomes & Impact

01 Quantifiable Outcomes

Data-driven metrics tracking tangible progress and accountability in our accessibility initiative.

Defect Reduction Measurable drop in unresolved accessibility defects, aligning with global WCAG compliance benchmarks.	Higher Satisfaction Elevated student satisfaction scores reflecting improved barrier-free facilities and inclusive experiences.
Diverse Participation Expanding cohort of cross-discipline participants fostering a holistic approach to inclusion.	AI Risk Mapping AI-driven identification of high-risk zones, enabling proactive and targeted accessibility upgrades.

02 Learning Outcomes

Building essential competencies and fostering a culture of empathy and collaboration across the campus community.

Inclusion Literacy Universal understanding of accessibility principles across the student body, breaking down awareness barriers.	Needs Identification Enhanced ability to recognize and articulate unmet accessibility needs in physical and digital realms.
Feedback Mastery Proficiency in submitting structured, actionable feedback that drives meaningful institutional change.	Collaborative Skills Hands-on experience in cross-disciplinary teamwork, solving complex accessibility challenges together.

Budget Overview — Total HK\$100,000



GRAND TOTAL ALLOCATED

HK\$100,000

Funding is allocated across tech infrastructure, marketing, production, and advanced AI analytics to ensure project success.

Note: AI application is restricted to structured historical facility data only, strictly following panel guidelines.

Team & Stakeholders

01 Core Team



Tsoi Pui Yan

Team Leader

ESE Year 3 | Oversees project strategy, timeline management and stakeholder coordination.



Chong Wai Ting

Data & Analytics

DS Year 2 | Analyzes survey data, compiles reports and visualizes project metrics.



Liu Yingge

Communications Lead

ESE Year 3 | Manages internal/external communications and community engagement.

02 Key Partners & Stakeholders

FMO (Facilities Mgmt)

Provides infrastructure support, venue access and logistical coordination for on-campus events.

Central IT Services

Supports digital tools, network access and provides technical troubleshooting for the team.

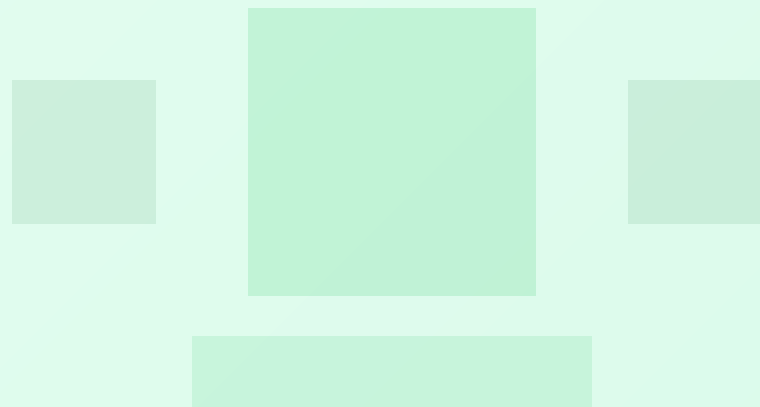
SDS (Student Dev)

Offers guidance on student engagement programs, leadership training and activity funding.

Sustainability Club— A key student partner for on-ground promotion, volunteer recruitment and aligning projects with campus green goals.

Academic Faculties— Faculty advisors from Engineering & Science provide academic oversight, research validation and professional mentorship.

Thank You



Building a more inclusive CityUHK, one QR code at a time
